

Important Terminologies in Probability

1. Random Experiment

Definition

A **random experiment** is an activity or process that has **more than one possible outcome**, and **the exact outcome cannot be predicted in advance**, even though all possible outcomes are known.

In simple words:

A random experiment is an experiment whose result is uncertain before it is performed.

Key Characteristics

A random experiment has the following properties:

- It can be repeated under similar conditions.
- There are multiple possible outcomes.
- The exact outcome is uncertain.
- The set of all possible outcomes is known beforehand.

Examples

Example 1: Tossing a Coin

Possible outcomes:

- Head (H)
- Tail (T)

Before tossing, we know the possible outcomes, but we cannot predict which one will occur.

Therefore, tossing a coin is a random experiment.

Example 2: Rolling a Die

Possible outcomes:

1, 2, 3, 4, 5, 6

We know all possible outcomes but cannot predict the exact number before rolling.

Example 3: Drawing a Card

Possible outcomes:

52 different cards

We know the possible cards, but not which one will be drawn.

Example 4: Weather Forecast

Will it rain tomorrow?

Possible outcomes:

- Rain
- No Rain

Although meteorologists estimate the probability, the exact outcome is uncertain.

Non-Random Experiment Examples

Example 1

What is $2 + 3$?

The answer is always 5.

This is **not** a random experiment.

Example 2

Switching on a working light.

If electricity is available, the bulb glows every time.

This is deterministic, not random.

AIML Connection

Machine Learning models deal with uncertainty.

For example:

- Will a customer buy a product?
- Will a patient have diabetes?
- Will an email be spam?

These are treated as random experiments because the outcome is uncertain.

2. Trial

Definition

A **trial** is **one performance** of a random experiment.

Every time the experiment is conducted, it is called a trial.

Example 1

Coin Toss

Experiment:

Toss a coin.

One toss = One trial

Important Point

Random Experiment = Overall activity

Trial = One execution of that activity

Analogy

Experiment:

Roll a die.

Trials:

- First roll
- Second roll
- Third roll

3. Outcome

Definition

An **outcome** is the result obtained after performing a trial.

Example 1

Coin Toss

Possible outcomes:

Head

Tail

If the coin shows Head,

Head is the outcome.

Key Points

One trial produces exactly one outcome.

Different trials may produce different outcomes.

4. Sample Space

Definition

The **sample space** is the **set of all possible outcomes** of a random experiment.

It is represented by **S**.

Importance of Sample Space

Probability is always calculated using the sample space.

Formula

$$P(E) = \text{Favourable Outcomes} / \text{Total Outcomes}$$

The denominator (Total Outcomes) comes from the sample space.

5. Event

Definition

An **event** is a **set of one or more outcomes** from the sample space that satisfies a particular condition.

An event is always a subset of the sample space.

Example 1

Die Roll

Sample Space

$$S = \{1, 2, 3, 4, 5, 6\}$$

Event

Getting an even number

$E = \{2, 4, 6\}$

Real-Life AIML Example

Suppose a bank wants to predict whether a customer will default on a loan.

- **Random Experiment:** Predicting whether a customer defaults.
- **Trial:** One customer's loan application.
- **Outcome:** Default or No Default.
- **Sample Space:** {Default, No Default}
- **Event:** Customer defaults.

This is exactly how probability concepts are applied in Machine Learning classification problems.

Types of Events

1. Simple Event

Contains only one outcome.

Example

Rolling a die

Getting 4

Event

$\{4\}$

2. Compound Event

Contains more than one outcome.

Example

Getting an even number

$\{2,4,6\}$

3. Impossible Event

Cannot occur.

Example

Getting 7 on a standard die

$\emptyset$ (Empty Set)

Probability = 0

4. Certain Event

Always occurs.

Example

Getting a number less than 7 on a die.

Event

$\{1,2,3,4,5,6\}$

Probability = 1

Rules of Probability

Probability follows certain mathematical rules that help us calculate the likelihood of events. Understanding these rules is essential for Machine Learning, Statistics, and Data Science.

1. Probability Value Rule

Rule

The probability of any event always lies between **0 and 1**.

$$0 \leq P(A) \leq 1$$

Meaning

- **0** → Impossible event
- **1** → Certain event
- Between **0 and 1** → Event may or may not happen

2. Total Probability Rule

Rule

The probability of the entire sample space is always **1**.

$$P(S)=1$$

where **S** represents the sample space.

Example

Rolling a die

Sample Space

$$S=\{1,2,3,4,5,6\}$$

Probability

$$\frac{1}{6} + \frac{1}{6} + \frac{1}{6} + \frac{1}{6} + \frac{1}{6} + \frac{1}{6} = 1$$

3. Impossible Event Rule

Rule

An impossible event has a probability of **0**.

$$P(\emptyset) = 0$$

where $\emptyset$ (empty set) represents an impossible event.

Example

Rolling a standard die

Event:

Getting 8

Impossible

Probability

$P(8)=0$

4. Certain Event Rule

Rule

A certain event always has a probability of 1.

Example

Rolling a die

Event:

Getting a number less than 7

Possible outcomes

1,2,3,4,5,6

Probability

$$\frac{6}{6} = 1$$

5. Complement Rule

Rule

The probability that an event does not occur is called its complement.

Formula

$$P(A') = 1 - P(A)$$

where

- A = Event occurs
- A' = Event does not occur

Example 1

Probability of getting a Head

$$P(H) = \frac{1}{2}$$

Probability of not getting Head

$$P(H') = 1 - \frac{1}{2} = \frac{1}{2}$$

6. Addition Rule (Union Rule)

The addition rule is used when we need the probability of A OR B occurring.

Formula

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Why subtract the intersection?

Because the common part ($A \cap B$) gets counted twice.

Example

A class has 100 students.

- Python = 40
- SQL = 30
- Both = 10

Find the probability that a student knows **Python or SQL**.

$$P(\text{Python} \cup \text{SQL}) = \frac{40+30-10}{100} = \frac{60}{100} = 0.6$$

7. Multiplication Rule (Intersection Rule)

Used to calculate the probability of **A AND B** occurring together.

For Independent Events

If the occurrence of one event does not affect the other,

$$P(A \cap B) = P(A) \times P(B)$$

Example

Coin Toss and Die Roll

Probability of getting

- Head
- 6

$$P(H) = \frac{1}{2}$$

$$P(6) = \frac{1}{6}$$

$$P(H \cap 6) = \frac{1}{2} \times \frac{1}{6} = \frac{1}{12}$$

For Dependent Events

If one event affects the other,

$$P(A \cap B) = P(A) \times P(B | A)$$

Example

A bag contains:

- 3 Red Balls
- 2 Blue Balls

Two balls are drawn **without replacement**.

Find the probability of drawing two red balls.

First Red

$$P(R1) = \frac{3}{5}$$

Second Red (after one red is removed)

$$P(R2 | R1) = \frac{2}{4}$$

Final probability

$$\frac{3}{5} \times \frac{2}{4} = \frac{6}{20} = \frac{3}{10}$$

8. Mutually Exclusive Events Rule

Two events are **mutually exclusive** if they cannot happen at the same time.

Formula

$$P(A \cap B) = 0$$

Therefore,

$$P(A \cup B) = P(A) + P(B)$$

Example

Rolling a die

- Event A = Getting 2
- Event B = Getting 5

A die cannot show both 2 and 5 at the same time.

So,

$$P(A \cap B) = 0$$

9. Independent Events Rule

Two events are **independent** if the occurrence of one does not affect the other.

Formula

$$P(A \cap B) = P(A) \times P(B)$$

Example

- Tossing a coin
- Rolling a die

Getting Head does not affect getting 6.

